

UNIT-2: Database backup & CSV handling

2.1 SQLite dump:

- '.dump' is the command which allows user to dump entire database, table, structure etc. into text file.
- In another word, it is used to take backup of database into file.
- Example:


```
.open student
create table stud_info(rno, name, gender);
insert into stud_info values(1,'Reet','M'),(2,'Meet','M'),
(3,'Geet','F'),(4,'Jeet','M');
create table stud_result(rno, m1,m2,m3);
insert into stud_result(1,45,65,76),(2,54,66,87),(3,45,56,67),(4,78,89,87);
```
- Student database:
 - o stud_info & stud_result tables

2.1.1 Dump entire database into file

- '.dump' command is used to dump entire database into file.
- Syntax:


```
.dump
```

 : It is used to dump the whole database related information.
- By default '.dump' command dump content on output screen.
- To save output into specific file, '.output' command is used.
- Syntax:


```
.output filename
.dump
```
- Now the dump output is stored into given filename.
- This file is stored in SQLite folder.
- Example:


```
.output temp.txt
.dump
```

```
PRAGMA foreign_keys=OFF;
BEGIN TRANSACTION;
CREATE TABLE stud_info(rno, name, gender);
INSERT INTO stud_info VALUES(1,'Reet','M');
INSERT INTO stud_info VALUES(2,'Meet','M');
INSERT INTO stud_info VALUES(3,'Geet','F');
INSERT INTO stud_info VALUES(4,'Jeet','M');
CREATE TABLE stud_result(rno, m1,m2,m3);
INSERT INTO stud_result VALUES(1,45,65,76);
INSERT INTO stud_result VALUES(2,54,66,87);
INSERT INTO stud_result VALUES(3,45,56,67);
INSERT INTO stud_result VALUES(4,78,89,87);
COMMIT;
```

2.1.2 Dump specific table into file

- To dump specific table into file DUMP command is used with table name.
- Syntax:


```
.output filename
.dump table_name
```

- Example:

```
.output temp.txt
.dump stud_info
```

```
PRAGMA foreign_keys=OFF;
BEGIN TRANSACTION;
CREATE TABLE stud_info(rno, name, gender);
INSERT INTO stud_info VALUES(1,'Reet','M');
INSERT INTO stud_info VALUES(2,'Meet','M');
INSERT INTO stud_info VALUES(3,'Geet','F');
INSERT INTO stud_info VALUES(4,'Jeet','M');
COMMIT;
```

- The information dumps into the file till you quit from SQLite mode.

- Example:

```
Select * from stud_result;
```

```
PRAGMA foreign_keys=OFF;
BEGIN TRANSACTION;
CREATE TABLE stud_info(rno, name, gender);
INSERT INTO stud_info VALUES(1,'Reet','M');
INSERT INTO stud_info VALUES(2,'Meet','M');
INSERT INTO stud_info VALUES(3,'Geet','F');
INSERT INTO stud_info VALUES(4,'Jeet','M');
COMMIT;
1|45|65|76
2|54|66|87
3|45|56|67
4|78|89|87
```

- To stop dumping output into file, use '.quit' command.

2.1.3 Dump only table structure

- To view the table structure, '.schema' command is used.
- To dump table structure into file again use '.output' command.
- Syntax:

```
.output filename
.schema tablename
```

- Example:

```
.output temp.txt
.schema stud_info
```

- It dumps the structure of stud_info into temp.txt file.

2.1.4 Dump data of one or more tables into a file

- To dump table data of one or more tables into file, first change mode setting.
.mode insert (It displays insert statements of each rows)
- Write select statements to display each rows' data into table.

- Example:

```
.mode insert
.output temp.txt
Select * from stud_info
```

```
INSERT INTO "table" VALUES(1,'Reet','M');
INSERT INTO "table" VALUES(2,'Meet','M');
INSERT INTO "table" VALUES(3,'Geet','F');
INSERT INTO "table" VALUES(4,'Jeet','M');
```

2.2 CSV files handling:

- CSV stands for "Comma Separated Value".
- Stud.csv file

```
Rollno,Name,Address,Gender
101,Smit,Varachha,M
102,Jasmeet,Katargam,M
103,Jenish,Amroli,M
201,Jaimini,Varachha,F
202,Tanshukh,Katargam,M
203,Uma,Ved road,F
```

2.2.1 Import a CSV file into a table

- CSV file can be imported using SQLite.
- Syntax:
 - .mode csv
 - .import csv_filename tablename
- Example:
 - .mode csv
 - .import stud.csv stud
- These commands import stud.csv file into stud table.
- SQLite consider first line of CSV file as heading for column and all other lines are considers as records of table.
- To see the structure of STUD table:

```
.schema stud
```

```
CREATE TABLE IF NOT EXISTS "stud"(
  "Rollno" TEXT,
  "Name" TEXT,
  "Address" TEXT,
  "Gender" TEXT
```

- To see records of STUD table:

```
Select * from stud;
```

```
101,Smit,Varachha,M
102,Jasmeet,Katargam,M
103,Jenish,Amroli,M
201,Jaimini,Varachha,F
202,Tanshukh,Katargam,M
203,Uma,"Ved road",F
```

2.2.2 Export a CSV file from table

- Table content can be exported to CSV file using SQLite.
- Write following code:
 - .mode csv
 - .header on
 - .output temp.csv
 - Select * from stud;
- '.mode csv': converts the mode to csv format.

- '.header on': display column heading on top (means in first line).
- 'output temp.csv' : allow to redirect results into temp.csv file.
- 'select * from stud': it displays tables' record related information.
- Output:

```
Rollno,Name,Address,Gender  
101,Smit,Varachha,M  
102,Jasmeet,Katargam,M  
103,Jenish,Amroli,M  
201,Jaimini,Varachha,F  
202,Tanshukh,Katargam,M  
203,Uma,Ved road,F
```